Team InfraWarriors

LOG BOOK

Autonomous simulation-based optimized path finding Robot

PeraBots'25 Elimination Round - University Category

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As a team of engineering students from the Faculty of Engineering, University of Moratuwa, we are participating in the PeraBots'25 competition to showcase our innovation in autonomous robotics.

1.1 Algorithmic Approaches

The primary objective of our robot is to navigate the simulated environment by identifying and following the **optimized path** while ensuring efficient **obstacle detection and avoidance**. The robot is designed to intelligently explore the given arena, evaluate possible routes and dynamically adapt its path to reach the target in **minimal time**. By combining **path optimization algorithms** with real-time **sensor-based decision making**, our robot aims to achieve both **speed and accuracy** in completing the task.

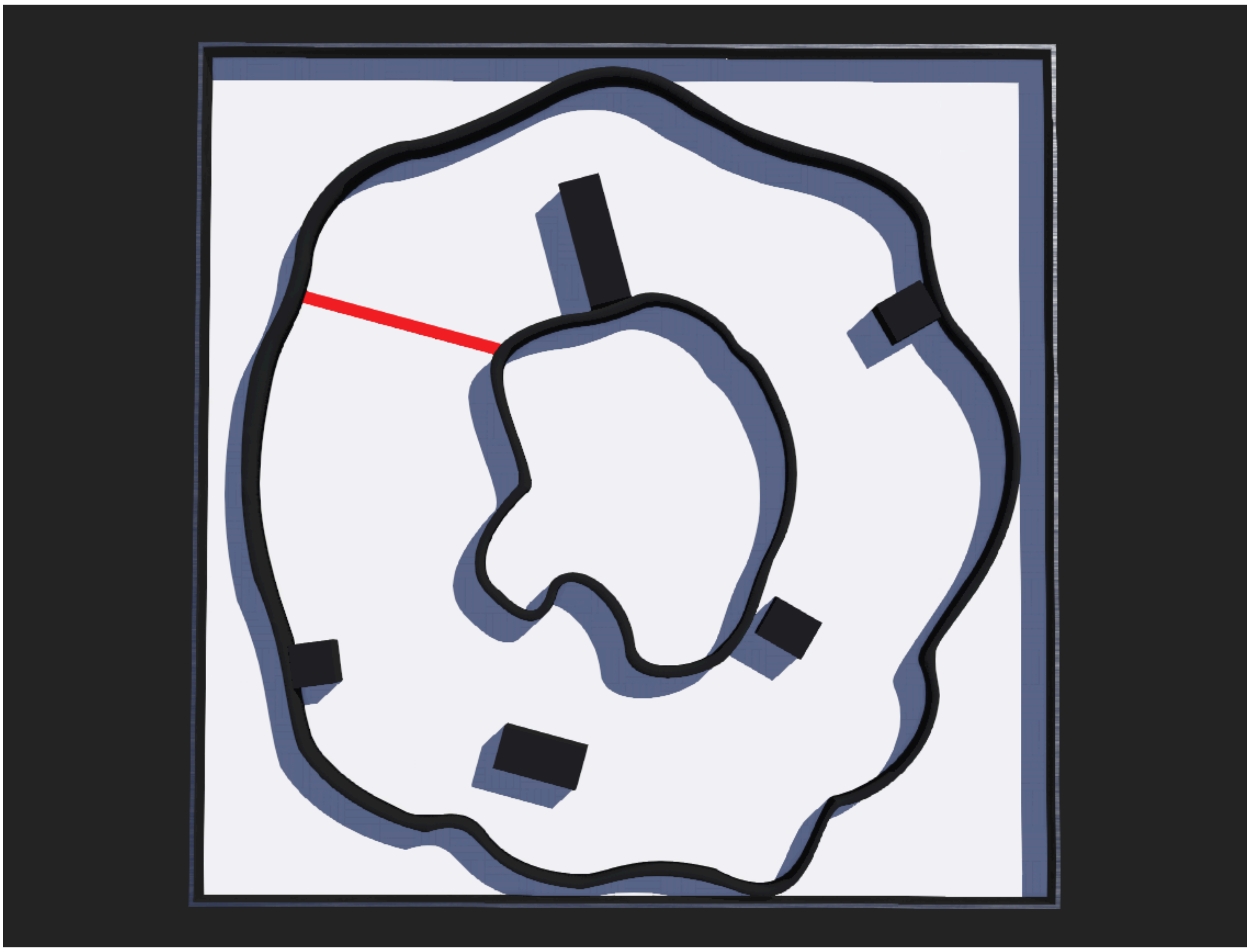


Fig. 1 - Sample Arena of the Task

- ## Obstacle Avoidance Concept

The robot uses a **LIDAR**-based dynamic obstacle avoidance system that operates by interpreting environmental feedback in real time. The robot divides the space around it into angular segments using **LIDAR** readings and then makes decisions based on distance data, allowing it to navigate around obstacles without external assistance.

Key Mechanism

- **LIDAR Segmentation**

  The 540-point LIDAR data is divided into 6 key sectors representing different directional zones: far-left, mid-left, front-left, front-right, mid-right, and far-right. These segments help the robot approximate the spatial location of nearby obstacles.

- **Distance Averaging**

  Each 90-point segment is averaged to determine the presence of nearby objects. If the average value in a sector falls below the defined **Turn threshold**, the robot considers that direction blocked.

- **Dynamic Navigation Logic**

  The robot moves forward if the central (front) sectors are clear. If not, it analyzes side sectors to determine the optimal turning direction. Turning angles are calculated relative to the last movement direction to reduce unnecessary rotation.

- **GPS-Supported Safety Zones**

  In tight or complex parts of the environment, the robot uses GPS location to adjust its sensitivity by altering the **Turn threshold** and **Front threshold**. This allows the robot to react more cautiously or aggressively depending on the zone (narrow curves, near obstacles or open paths).

Key Features

- Real-time dynamic adaptation based on sensor data.
- GPS-based situational behavior refinement.
- No collisions due to proactive checking.

• Path Correction

Condition	Action
Obstacle ahead	Turn left/right
Right clear	Prefer right curve
Obstacle on right	Curve to left / Proceed straight
Center path clear	Proceed straight
else	else

• Lap time optimization logic

Beyond merely avoiding obstacles, the robot aims to complete its lap in the shortest time possible. To achieve this, the controller implements various strategies that reduce delays, eliminate redundancy, and ensure efficient path following.

Optimization Strategies

- **Minimum Rotation Strategy**

  Using a direction reference (pos), the robot calculates how much to turn based on the difference between current and desired direction. This avoids excessive turning and ensures the robot takes the most direct path to realign

- **Position-Based Behavior Adaptation**

  The controller dynamically alters sensor thresholds based on the robot's position (from GPS) to behave differently in known sections of the map,

  - Narrow areas - Thresholds are lowered for more sensitivity.
  - Open spaces - Thresholds are increased to ignore minor, irrelevant LIDAR returns.

- **Checkpoint Handling**

  At specific points (corners or goal zones), the robot pauses, recalibrates or changes its logic to suit the situation. This prevents repeated unnecessary maneuvers and allows for a more optimized path through known checkpoints.

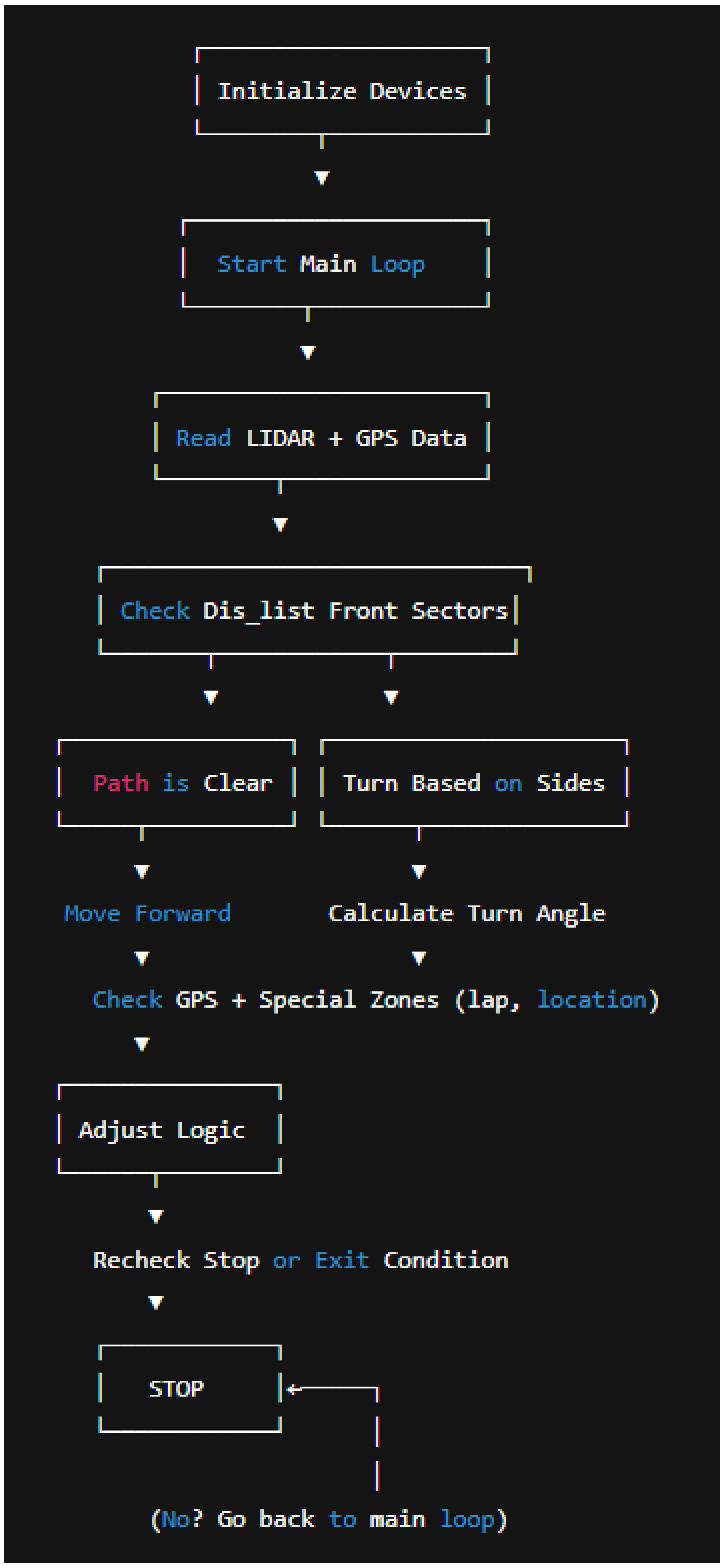
- **Redundant Path Avoidance**

  During multiple laps (tracked using lap variable), the robot modifies its strategy to skip earlier exploration logic. This ensures subsequent laps are executed more directly using previously learned orientation.

Effectiveness

- Avoids time wasted on indecisive navigation.
- Learns and adjusts to path structure through lap tracking.
- Combines real-time sensor interpretation with stored directional bias (pos) for better control.

- ## Flowchart



1.2 Development Process

- We tested our robot multiple times to improve its lap time. During these tests, we observed the robot's performance under different conditions.

- We discovered that maintaining the robot's angular velocity at **5.7 rad/s** produced the best results. When we deviated from this value, the robot tended to drift away from the intended path. This indicated that the robot's control algorithm and sensor response were optimized for that specific angular velocity, helping it stay centered on the line and maintain stability during turns.

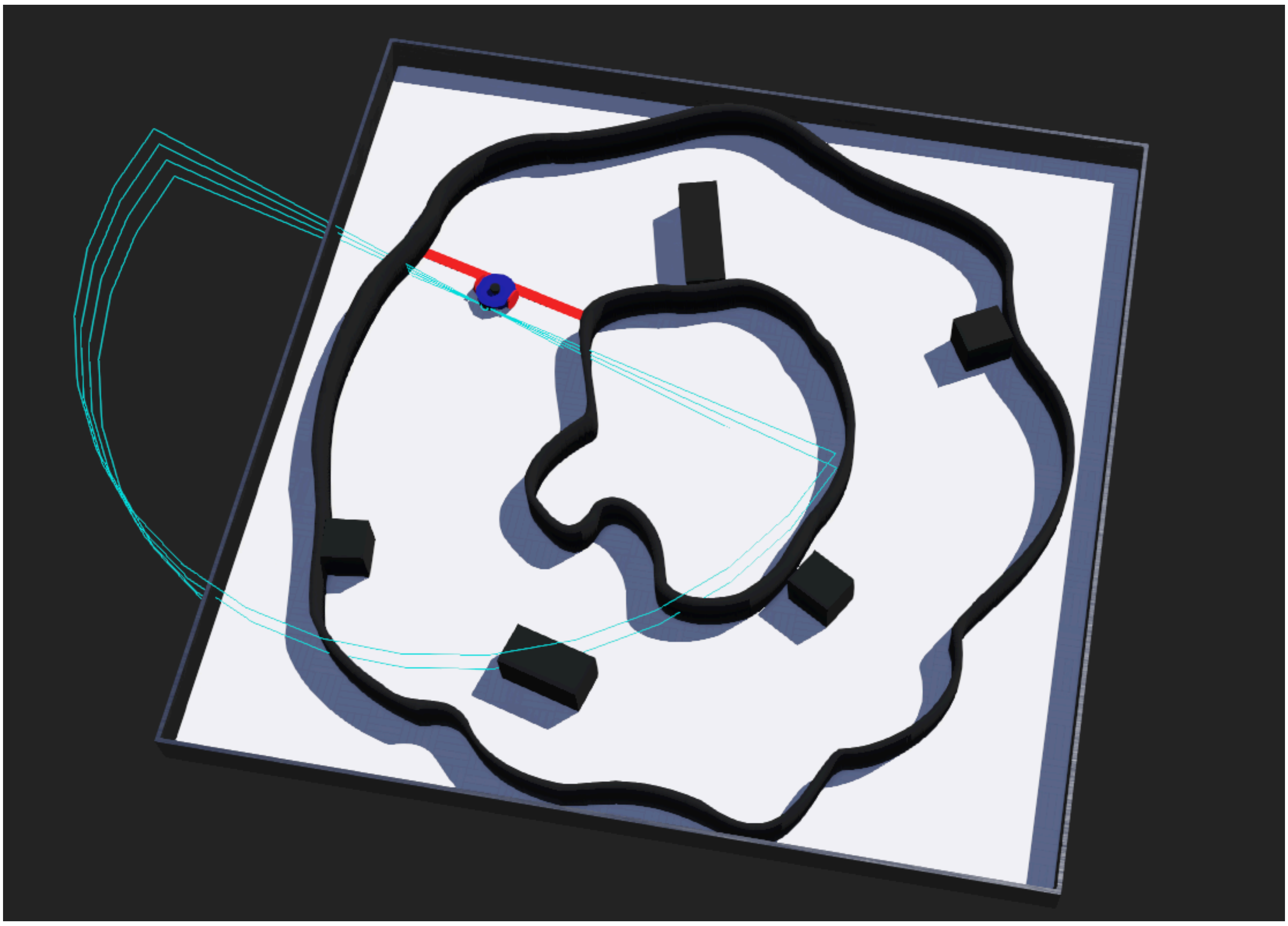


Fig. 2 - Sample Simulated Robot (Construction Phase)

- We also decided on the optimal rotation angle for the robot to turn when an obstacle is detected ahead.
- In addition, we determined the minimum safe distance between the robot and the obstacle to trigger the turning action.
- Through repeated testing, we identified the most suitable physical dimensions for our robot to perform this task efficiently, especially in tight spaces.
- Another key part of our development process was optimizing the **LiDAR sensor's range** for accurate obstacle detection. Fine-tuning the LiDAR's effective detection range was one of the main focuses during debugging, as it directly influenced the robot's ability to respond reliably and at the right time.

I.3 Challenges & Solutions

Developing an autonomous robot for the elimination round of PeraBots'25 competition presents a complex array of challenges, particularly in a simulation based environment. Navigating the intricacies of sensor accuracy, algorithm performance and system integration within a virtual setting required our team to employ creative problem solving strategies and iterative improvements. This section outlines the specific obstacles we encountered, the innovative solutions we developed and the enhancements that ultimately refined our robot's capabilities, showcasing our adaptability and commitment to excellence in autonomous robotics.

Challenges	Solutions
Understanding how the robot can detect obstacles and identify free space for navigation	• Implemented a LiDAR system to detect surroundings. • Generated a 2D map of the robot's environment. • Used a 180-degree LiDAR scan to continuously monitor for obstacles. • Continuously analyzed the map to identify the presence and absence of obstacles.
Determining how the robot should turn when an obstacle is detected (angle and direction)	• Divided the 180-degree scan into 6 sections, each spanning 30 degrees. • When an obstacle is detected in a section, the robot searches adjacent sections for clear paths. • Chooses the nearest obstacle-free section to determine the optimal turning direction and angle. • Robot turns towards the selected direction to avoid the obstacle and continue navigation.
Robot took too much time to identify obstacles	• During debugging, we tested and tuned the most optimized threshold values to reduce detection time. • Improved sensor response and filtering to enhance object identification speed.
Improving lap time	• Turns with minimal angle change. • Adjusts behavior based on location. • Optimizes actions at known checkpoints. • Skips redundant logic in later laps.
Deciding the dimensions of the robot	• Identified the minimum space the robot needs to pass through. • Based on this, finalized the optimal dimensions for smooth navigation.

Reflecting on the challenges in developing our autonomous robot within a simulation-based environment, we've grown significantly. Each problem solved enhanced our robot and strengthened our team. We're proud of our progress and confident in our readiness for the next stages of PeraBots’25

1.4 Innovations

- ## Sensor behavior

Sensor	Type	Data Used	Purpose
lidar	LIDAR (2D Laser Range)	540-point scan, grouped into 6 sectors	• Detects obstacles ahead and to sides • Sectors 0–5 (left to right) averaged every 90 points • Key for obstacle avoidance and determining clear directions
global	GPS	X, Y, Z coordinates	• Tracks current robot position • Used for custom zone detection and determining if robot reached specific points (turns, finish area, or checkpoints)
rotational_motor_1	Actuator (Left Motor)	Speed/Position	• Controls forward, backward, and turning motions
rotational_motor_2	Actuator (Right Motor)	Speed/Position	• Works together with left motor to handle precise turns and forward movement

- ## LIDAR Sector Mapping

LIDAR Sector	Index Range	Direction Covered
Sector 0	1–90	Hard Left
Sector 1	91–180	Slight Left
Sector 2	181–270	Front Left
Sector 3	271–360	Front Right
Sector 4	361–450	Slight Right
Sector 5	451–540	Hard Right

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